

Final Project Blueprint

Step	Action	Logic
1. Software Setup	Install Arduino IDE . Download the LiquidCrystal_I2C and Keypad libraries.	The compiler uses #include directives to bring in pre-written C code, allowing you to interface with complex hardware using simple Functions .
2. Electronics Connection (Inputs)	HC-SR04 Sensor: VCC to 5V, GND to GND, Trig to Pin A0, Echo to Pin A1. Keypad: Connect the 8 pins to Digital Pins 2 through 9. Push Button: Connect one leg to Pin 10, the other to GND.	Variables & Initialization: Assign physical connections to variables (const int trigPin = A0;). The button uses internal pull-up logic (INPUT_PULLUP), requiring no external resistor.
3. Electronics Connection (Outputs)	I2C LCD: VCC to 5V, GND to GND, SDA to Pin A4, SCL to Pin A5. Buzzer: Positive leg to Pin 11, negative to GND. Single LED: Cathode (short) to GND, Anode (long) via 220Ω resistor to Pin 13.	Pin Modes: In the setup() function, configure registers to define whether a pin sends voltage out (Trig, LED, Buzzer) or listens for it (Echo, Button).
4. State Machine Setup	Define the operating states of the system: inputting data, lifting, and resting.	Decision: An enum defines the states, and a switch(currentState) statement directs the microprocessor to execute only the code relevant to

		the current system state.
5. STATE_INPUT	Ensure the LED is OFF . Prompt the user on the LCD. Read keypad inputs to store the target reps and rest time. Wait for the Start Button to be pressed.	Function: A custom readKeypadInput() function parses characters into integers. Once the button is pressed, the state machine transitions to STATE_LIFTING.
6. STATE_LIFTING	Ensure the LED is ON . Trigger a 10μs sound pulse from the HC-SR04 and measure the echo return time. Calculate distance. If distance drops below threshold (e.g., 25 cm), count a rep.	Array & Logging: Check distance mathematically. If valid, increment rep counter, beep, and update LCD. Once target reps are reached, save elapsed time into the setTimes[] array and transition to STATE_RESTING.
7. STATE_RESTING	Ensure the LED is OFF . Display a countdown timer on the LCD based on the inputted rest time.	Loop: A while loop runs continuously, subtracting elapsed time from the target rest time. It updates the LCD every second without blocking the Start/Stop emergency reset button.
8. Alarm & Reset	Once the rest timer hits zero, emit a long beep from the piezo buzzer and return to STATE_LIFTING for the next set.	Function: A custom triggerRestAlarm() function controls the buzzer. The system repeats until the user manually stops the workout.

The Final C Code Implementation